

perturbation

Llama-3.1-8B-It
+ OCT LoRA, $r=64$
4 constitutions
× 4 dose rungs



CAA extraction

8 traits ×
10 personas
~500 A/B items
activation at the
answer token
layers 15, 20



trait vectors

$$V_{c,t,p} = \text{mean}(\text{pos}) \\ - \text{mean}(\text{neg})$$

$$dV_{c,t,p} = \\ V_{c,t,p} - V_{t,p}^{\text{base}}$$



statistics

$$dG_{c,t} = \text{mean}_p dV$$

fig 1 size, fig 5 sign

dispersion, RDM
fig 3 dose, fig 4 control

$$\mu + C + T + P + CT + CP + TP + CTP$$

fig 2

cross-fitted on disjoint
question halves; question
bootstrap for all intervals

matched control



untrained LoRA:
iid / spectral /
permuted, sited
by MEASURED dose